

ILHAM MRABTI

GIS & Geospatial Data Consultant | Environmental Geomatics Specialist | PhD Researcher

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[linkedin.com/in/ilham-m-rabti-98b4a3232](https://www.linkedin.com/in/ilham-m-rabti-98b4a3232) | Portfolio: mrabtiGis.netlify.app | Open to freelance & remote projects

PROFESSIONAL SUMMARY

GIS and geospatial data specialist with hands-on experience delivering spatial analysis, environmental impact studies, and geodatabase management for engineering and environmental consulting projects. Skilled in GIS mapping, PostgreSQL/PostGIS database administration, remote sensing, and environmental modeling (SWAT, InVEST). Currently a PhD researcher in Geomatics, combining applied technical expertise with a strong scientific and analytical foundation. Detail-oriented and reliable, turning raw spatial data into clear, decision-ready maps, reports, and databases — delivered accurately and on schedule.

SERVICES

- GIS mapping & thematic cartography for technical and regulatory reports
- Spatial database design & administration (PostgreSQL/PostGIS)
- Spatial data collection, analysis & synthesis for environmental impact assessments
- Remote sensing image processing & analysis
- Hydrological & environmental modeling (SWAT, InVEST)
- Spatial data quality control, topology checks & multi-source data integration

CORE SKILLS

- GIS & Geospatial Analysis: ArcGIS Pro, ArcMap, QGIS, Google Earth Engine (GEE), ENVI, Global Mapper., TerrSet; multi-criteria overlay analysis, spatial queries, spatial joins & intersect analysis, geometric processing
- Databases: PostgreSQL/PostGIS — design, administration, quality control, multi-source integration
- Remote Sensing: Sentinel-2 and Landsat 8 image processing, change detection analysis
- Environmental Modeling: SWAT / SWAT+ and InVEST hydrological/environmental models
- Cartography & Mapping: technical and thematic map production to industry standards
- Data Preparation: georeferencing, mosaicking, digitizing, CAD-to-GIS conversion (CAD/CAO to GeoJSON)
- Quality Assurance: topology verification, error detection, data validation

SELECTED PROJECTS

Post-Fire Burned Area Assessment Chefchaouen Forest Region, Morocco

- Assessed wildfire-affected forest zones using multi-temporal Sentinel satellite imagery and change detection techniques.
- Compared pre- and post-fire imagery to delineate and quantify burned areas.
- Produced thematic maps of fire extent to support environmental assessment and reporting.
- Tools: Sentinel-2 imagery, remote sensing change detection, QGIS

Cadastral Road Realignment & Impacted-Owner Analysis Morocco

- Georeferenced and mosaicked 30+ scanned cadastral PDF plans into a unified, accurate spatial dataset.
- Converted legacy CAD (CAO) plans into GeoJSON and cross-checked them against updated imagery to detect parcel changes, digitizing updates and assigning parcel IDs.
- Joined parcel geometries with property-owner records and ran spatial intersect analysis to identify owners affected by a proposed road realignment.
- Delivered a full GIS workflow from raw scanned plans to actionable stakeholder-impact analysis.
- Tools: georeferencing, mosaicking, CAD-to-GIS conversion, digitizing, spatial joins, intersect analysis

Land Use/Land Cover Impact on Water Balance Environmental Assessment Studies, Morocco

- Used Landsat 8 imagery and the SWAT+ hydrological model to assess the impact of land use/land cover change on water balance components.
- Produced full cartographic deliverables for each study, including situation maps, geological maps, hydrographic network maps, and piezometric maps.
- Tools: Landsat 8, CHIRPS, ERA5, SWAT+, QGIS+, SWAT+ Toolbox

PROFESSIONAL EXPERIENCE

Environmental Geomatics Specialist YOUNAMEZ

Oct 2023 – Aug 2025 | Beni Mellal, Morocco

- Supported environmental impact assessments by collecting, analyzing, and synthesizing spatial data for regulatory submissions.
- Administered PostgreSQL/PostGIS geodatabases end-to-end: structuring, quality control, integration, and updates of spatial datasets.
- Ran advanced spatial analyses (multi-criteria overlays, spatial queries, geometric processing) to inform planning and decision-making.

- Produced technical and thematic maps to consulting-industry standards, integrated into engineering reports and project deliverables.

GIS 3D/BIM Intern SIGMATOP

Oct 2022 – Jan 2023 | Témara, Morocco

- Built a complete geodatabase for the VSB housing operation in Kénitra, structuring multi-source data into a unified system.
- Digitized land parcels and cadastral records to update cadastral layers.
- Performed quality control on geographic data, including topology verification and inconsistency detection prior to final integration.

EDUCATION

PhD Candidate, Geomatics In progress

Geomatics, Georesources and Environment Laboratory, Faculty of Sciences and Techniques, Sultan Moulay Slimane University, Béni Mellal, Morocco.

Master's Degree, Environmental Geomatics FST Beni Mellal

Sep 2020 – Jun 2022 | Beni Mellal, Morocco

Bachelor's Degree, Geomatics & Land Management FST Beni Mellal

Sep 2017 – Jun 2019 | Beni Mellal, Morocco

LANGUAGES

- Arabic — Native
- Amazigh — Native
- French — Fluent (B2)
- English — Professional working proficiency (B1)